

Module-3

Sampling: Concepts, Types of Sampling, **Probability Sampling:** simple random sampling, systematic sampling, stratified random sampling, cluster sampling,

Non Probability Sampling: convenience sampling- judgmental sampling, snowball sampling, quota sampling, Errors in sampling.

Sampling: Concepts

What is Sampling?

Sampling is a process in which the fixed numbers of observations are taken randomly from a larger population. A technique which is fundamental for behavioral research is known as sampling and without using it, research work is not possible. It is impracticable and Impossible to study the whole population due to practical limitations of cost, time and other factors that are indispensable and operative in studying the whole population. For the sole reason of making the research findings economical and accurate, the concept of sampling has been introduced.

For example, for taking decision about to purchase or not to purchase the fruits, a fruit merchant inspects randomly only a few of them rather than examining each and every fruit. The important objective of sampling is to obtain maximum information of the population under study using minimum of money, labour and time.

Sampling in Research:

A definite plan to obtain a sample from the sampling frame is sample design. The method or technique which is adopted by the researcher in selecting the units of sampling from the population is called sampling design.

The basis for selecting a sample survey is the framework or road-map which is called sample design and it affects other important aspects of the survey. For obtaining some type of relevant information using survey, researchers execute it for some population, or universe. The sampling frame must be defined in such a way that it represents the population of interest out of which a sample is to be drawn at random. This sampling frame should be either identical to the population or may be a part of it. The sampling frame can be indirectly related to the population subject to some under coverage (e.g., the population is preschool children and the frame is a listing of preschools). The design of the sample may be simple or complex which provides the fundamental plan and methodology for the selection of the sample.

Definition of Sampling

According to Davis S. Fox :

"In the social science, it is not possible to collect data from every respondent relevant to our study but only from some fractional part is called sampling".

According to Cochran :

"In every branch of science we lack the resources, to study more than a fragment of the phenomena that might advance our knowledge". In this definition a fragment is the sample and phenomena is the population.

Basic Concepts of Sampling

Following are the basic concepts of sampling in research methodology :

1) Universe/Population :

The entire group of units, being the focus of study, is represented by the population or universe consisting of all the persons in the country, in a particular geographical location, or a special domestic or economic group based on the purpose and scope of study. Non-human entities like farms, houses or business establishments also comprise population.

For example, if a study is to be conducted on the marks obtained by the students of Statistics then the 'universe' will be all the students of that subject in that class. If the class is consisting of 50 students then the 50 students is the "universe".

"Population or universe" is the sum total of the objects under study, both animated and in animated. Thus, numerical details of the collection of the individuals or of their qualities, or of results of the operations are known as universe. Universe can be classified as follows :

i) Finite Universe :

A 'finite universe' comprises of finite number of entities or members. For example, the universe of the heights of students in a particular class.

ii) Infinite Universe :

If the infinite numbers of members are included in a universe then it is called "infinite universe". For example, the universe of pressures at various points in the atmosphere.

2) Statistical Population :

To draw out statistical inferences from the set of entities of statistical population is based on random sampling of the population. For example, to derive generalizations about the crows, to only the set of crows should be the interest of study. However, if one chooses a population such as 'all crows', then the limitations will be in observing crows both the existing as well as existing in the future. There will be additional limitations of geography where the resources are limited for studying crows.

The total of quantities that are measurable or set of numbers is 'statistical population'. Thus, when only a single character symbolized every element of such a set then it is defined as 'single population', c.g., income of individuals. Depending on the finite and infinite number of elements, statistical population can be infinite as well as finite. A random set may not be a statistical population.

For example, a set of crows at a point of time will not represent a statistical population.

3) Sample :

A part of the population which is examined to estimate the characteristics of the population is called a sample.

For example,

- A portion of rice is examined to assess the quality of rice in the bag where the portion of rice examined is the sample and the bag of rice is the population.
- From a large consignment, proportion of defective articles is estimated only from a portion of the consignment. The selected portion is called sample.

4) Census:

When in a research every item or entity of the universe is included, that approach is called "census". In this method, detailed information regarding every population unit is collected. It is also called complete enumeration. This method provides more accuracy in the results in comparison to the sampling method.

For example, population survey about various issues, such as, literacy, education, income, etc., include all the items of the population.

5) Sampling Frame:

The actual set of units and a list containing every member of the population from which a sample is drawn at random is called sample frame.

For example, if one wants to know the factors that are responsible for the patients being admitted in the hospital after an acute attack of asthma in a specified area, then it becomes necessary to know the names of all those people of that particular area being admitted into the hospital for that reason.

6) Sample Unit :

Every sample is made up of several members or components. These members or components are called "sampling units". A sampling unit may or may not be a sampling element. Generally sampling unit is referred as sampling element. But in case of sampling units consisting of several population units, each sampling unit is called a cluster of units. The individual population units which come under the cluster of units, are called sampling elements. Therefore in this case a sampling element is an individual object on which the research is done, and it is a component of sampling unit.

For example, in case of inspecting the quality of oranges, only few oranges are selected to examine and each selected orange is a sampling unit. But in case of finding, the literacy level of a particular colony, several houses are selected in that colony for research. These houses are sampling units. However the individual members of each house are approached for examining the literacy level, thus these individual members of several houses are sampling elements.

7) Sample Design :

Designing a plan for effectively drawing out a sample from the sampling frame is called 'sample design'. In this different tools or techniques are used by researchers for drawing sampling units from the sampling frame.

Characteristics of a Good Sample

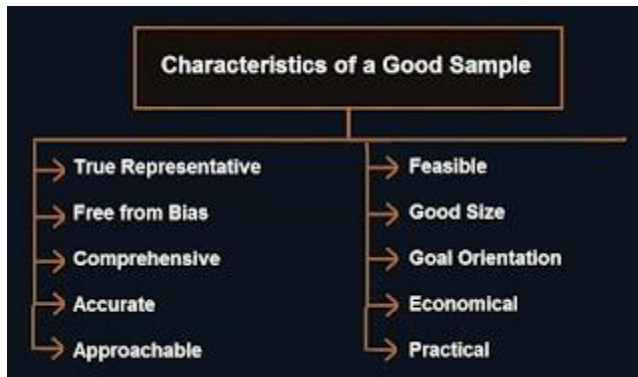
The various features of a good sample design are as follows :

A good sampling frame should be :

- Relevant : Research topic must be directly linked with relevant things.
- Complete : Coverage of all relevant items.
- Precise : It should exclude all the irrelevant items.
- Up-to-Date : Incorporating recent changes and additions, and deleting redundant items.

Characteristics of a Good Sample

The various features of a good sample design are as follows :



1) True Representative : The true representative of the population and matching its properties is termed as good sample where aggregate of certain properties is the population and sample is the sub-total of the universe.

2) Free from Bias : A good sample does not allow prejudices, pre-conceptions, and imaginations which affect its choice and it is unbiased.

3) Accurate : A sample is called good when it yields accurate estimates or statistics and free from errors.

4) Comprehensive : A sample that is true representative of the population is also comprehensive in nature which is controlled by definite purpose of investigation. A sample's characteristic may be comprehensive but it may not be a good representative of the population.

5) Approachable : The subjects of good sample are easily accessible where the tools of research are easily conducted and easy collection of data possible.

6) Good Size : The size of good sample is such that it yields an accurate result and the error due to probability can be estimated.

7) Feasible : A good sample creates the research work more feasible.

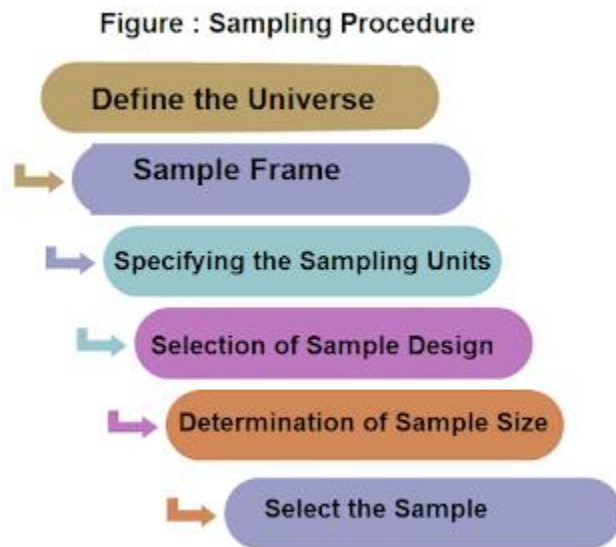
8) Goal Orientation : Any sample which is selected by the researcher should be able to satisfy the objectives of the research. The sample should be taken in proper number. It should be customized to fit the environment under which the research is going to be conducted. If the sample is changed as per the requirement of the survey design then it can come out with better results and outcomes.

9) Practical : It means that the concepts of sample selection should be applied properly while conducting the research. The researcher should be well experienced and well instructed. The instructions which are passed to observer should be clear, complete and correct in all terms so as to avoid errors and biasness on their part. The sample should be selected on basis of the sample design. The sampling units should be representative. The sample design should be practical and feasible in nature.

10) Economical : It refers that the research should not incur huge cost, time or efforts. One of the objectives of any research is to complete the research with minimum effort, time, money and resources. The researcher calculates per unit cost for each respondent. The researcher should choose that sampling design which gives minimum per respondent cost and maximum accuracy. The funding agency is highly concerned with per unit cost and the corresponding accuracy.

Procedure of Sampling

A brief description of which measurements will be taken at what times, on which material, in what manner, and by whom, is termed as 'sampling plan'. The designing of the sampling plans should incorporate that data to be evaluated which is a representative sample of the parameters of interest and be answerable to all the questions as stated in the goals. Following are the steps involved in sampling procedure in research or developing a sampling plan :



1) Define the Universe : The universe can be restricted to some geographical limits, a particular type of product or some other constraints. Defining the universe is the initial problem procedures of sampling. The collection of elements or objects possessing information desired by the researcher regarding what inferences are to be made is termed as the target population or the universe. Precise definition of the target population is must as ambiguous definition will result in ineffective research and deceptive in meaning. Proper defining of the target population requires translating the problem definition precisely as what should and should not be included in the sample. The description of the target population should be on the basis of elements, sampling units, the period and time. Thus, in survey research element is like an object from which or about information is required. In which survey research, the elements are usually the respondents.

For example, consider a marketing research project in which researcher wants to take response of the consumer for a new brand of men's perfume. It has to decide that who should be included in the target population. It is the important question all the men included in the research or the men who used the perfume during the last month or men who are 17 years of age, or older. Should females be included, because some women buy colognes for their husbands?. Appropriate description of target population requires prior resolution of such types of questions.

2) Sample Frame : The frame is enlisted by some population that already exists or it is constructed for some study to be conducted by the researcher. After the specification of the population to be studied, the frame of the population is developed. Sampling frame consists of all the sampling units of the population which is built by the researcher for the subject of his study or

it consists of already existing list of the population. This frame does not consist of names only but it may include a permanent location, a boundary, an address, or the set of certain rules that describes a sampling unit.

A set of boundaries which is restricting the universe is termed as a frame that may be in the form of lists, indices, maps, directories, population records, electoral rolls, city tax rolls, enrollment of the students in an university, etc. This frame is very mandatory in marketing studies and every element of the population enlisted appears only once which constitutes a sample frame. For a good sampling frame it should be accurate, free from duplication, and conveniently available. For the better performance of sampling procedure, sampling should be essential for marketing research. The representation of the elements of the target populations sampling frame which consists of a list to identify the target population.

For example, firms in industry that are enlisted in association directory, telephone book, a mailing list purchased from a commercial organisation, a city directory, or a map.

3) Specifying the Sampling Units : A sampling unit is decided on the basis of the sampling frame which is the primary unit containing the sampling elements of the population like, city blocks, households, a business organisation, etc. The design of the project is the partial basis of the selection of the sampling unit. The units that are served as initial sampling are called as primary sampling units. The objective of the inquiry is the basis of the composition of one or more units of the population.

For example, for the assessment of the consumer response to a new line of lipsticks, Lakme wants to sample females who are over 18 years of age where it can be the possibility of sampling such females directly and the sampling unit would be the same as an element. Otherwise, households might be the sampling units which are sampled. All the females above 18 in the selected households would be interviewed where the sampling unit differs from the population element. The time period under consideration is the extent of the geographical boundaries and the time factor.

4) Selection of Sample Design : The procedure of selecting units in the sample involves two primary methods of sampling namely, probability and non-probability methods that can be further sub divided into specific methods of selection. The probability sample being one and the procedure involved in selecting the units has some specific chance of its inclusion in the sample. An arbitrary method is adopted in selecting the units which is not depending upon the chance for the non probability sample. The purpose of inquiry and the attitude or convenience of the investigators is the basis of this method.

Two decisions are involved in the selection of the sample design:

- To use probability or non-probability method of selection.
- Specific sample design to use in collecting the data.

The following considerations affect the choice of the researcher :

- Usage of the probability sampling for the evaluation of the sampling error.
- Probability sample should be used for ensuring randomness in the selection of the units.
- Non-probability sampling should be used in the absence of proper sample frame.

- Non-probability sampling should be used when the considerations of time and money are important.

To achieve the objectives of investigation in the best possible way sample design should be selected after the decision of the method of selection whether probability or non-probability method. Unconcerned about the design being chosen finally the researcher has to defend his design at the time of the final presentation of the study results.

5) Determination of Sample Size : The desired degree of accuracy in investigation is directly proportional to the size the sample which also depends upon the nature of the population and the method of selection. The ideal samples size in marketing research investigations depends upon the the type of series and the population size. There will be more degree of heterogeneity and larger units s will be drawn in the sample due to larger size of the population. Hence the size of the sample should be larger for being fully representative.

6) Select the Sample : Execution of the actual sampling process is called as selection of the sample and the real selection of the elements of sample requiring considerable amount of office and field work especially when the personal interviews are involved. Detailed specification of the sampling design is the require is requirement for the execution of the sampling process with respect to the implementation of the size of population, sampling frame, sampling unit, sampling technique and sample size. Sampling units and sample size. being the household's requirement is the operational definition of a household, procedure for those houses that are vacant and if nobody is at home then the execution of the call backs is mandatory. For all the sampling design, decisions information must be provided in detail.

Advantages of Sampling

Following are the importance of sampling in research :

- 1) Saves Time, Money and Effort :** The subjects involved in sampling are small in number which gives him less time to calculate, tabulate, present, analyse, and interpret and hence the researcher saves time, money and effort.
- 2) More Effective :** Due to the size of the sample being small than that of the population, the tiredness is reduced in collecting the information and investigator works more effectively.
- 3) Faster and Cheaper :** Due to small sample size the data collection, tabulation, presentation, analysis, and interpretation takes less time and it involves nominal expenses.
- 4) More Accurate :** Small errors are in sampling because small data are involved in collection, tabulation, presentation, analysis and interpretation.
- 5) Gives More Comprehensive Information :** Thorough investigation of the study is the result of small sample which provides more complete information as all the members of the population have equal chance to be included in the sample.

Disadvantages of Sampling

Following are the disadvantages of sampling :

- 1) Biased Selection :** Biased selection of the respondents may be included in sampling by the research worker.

2) Difficulty in Selection : It is very difficult to select a truly representative sample because a large number of factors obstructing the method of selecting good samples.

3) Specialized Knowledge Needed : Specialized knowledge is required in the sampling method where the investigator may commit serious mistakes due to its absence.

4) Problem of Cooperation : Due to scattered sample the subjects are uncooperative with the researcher.

5) Less Accuracy : When the higher standard of accuracy is expected then the sampling is not suitable.

6) Limited Nature : Due to small or dissimilar universe it is impossible to derive a representative sample where census study is the best possible substitute.

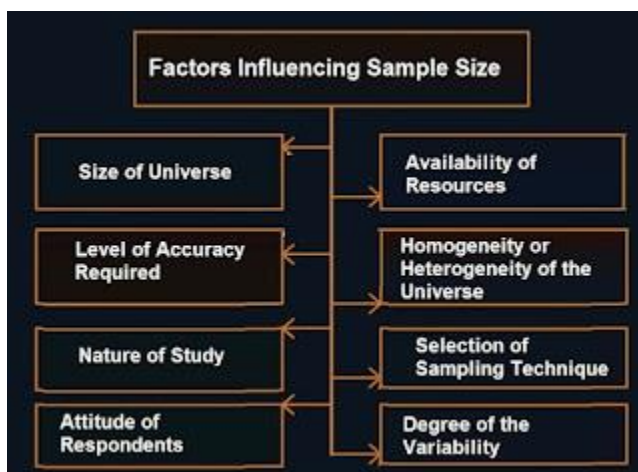
Sample Size

The sample size for a research refers to the total number of elements of the population to be included in the sample for conducting the research study. Both qualitative and quantitative points are involved in specifying the size of the sample for the research study. The accuracy of research depends on the size of the sample. It has been observed that a larger sample gives more accuracy and estimate levels. Apart from it, the availability of money, resources and efforts also specify the size of a sample.

The sample size of any research study is represented by 'n'. If the population is highly heterogeneous, then the sample will be divided into various groups with an indication that the sample size would be larger. If the population is homogeneous then the sample will be small. Thus, at pre-assigned level there is a fixation of the sampling error and the standard error of the statistic, thus having sample size as 'n' is chosen.

Factors Influencing Sample Size/ Sample Size Constraints

The following points should be taken care of while deciding the sample size :



1) Size of Universe : It has been observed and statistically proven that the sample size should be large so that it can represent the whole target population. If the universe size is large and heterogeneous then the researcher should take large sample size and vice versa.

2) Availability of Resources : The researcher needs a lot of resources to complete the task of research. These resources bind the researcher. It means that the researcher has to complete the research within those allotted resources. Resources can be money, time, experts or any other variable. If the resources are easily available then the sample size can be large otherwise smaller sample would be appropriate.

3) Level of Accuracy Required : For any research, the level of the accuracy affects sample size as bigger sample represents the whole population properly than a smaller sample. However, it is not always mandatory that a large sample will always give more accurate results. The selection of the sampling technique and research tools also impact the accuracy level of the research.

4) Homogeneity or Heterogeneity of the Universe : Homogeneous universe is that universe in which all the elements are similar, on the other hand heterogeneous universe is that universe where the elements are different from each other. A homogeneous universe requires a smaller sample size, while researcher needs to take bigger sample size in case of heterogeneous universe because a small sample may not represent the whole population.

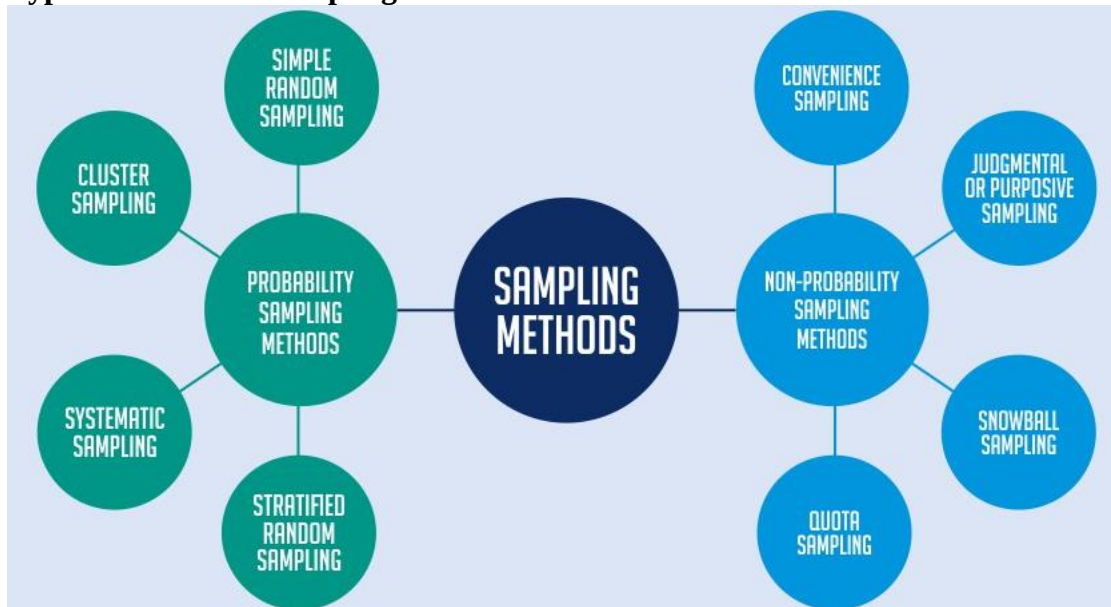
5) Nature of Study : The researcher may choose a small sample, if the research is intensive and fundamental (continuous) in nature. But in those cases, where the research is extensive and applied (researches which are not going to be repeated soon), then the researcher is bound to select larger sample size. So, it is quite clear that the nature of study also affects the sample size.

6) Selection of Sampling Technique : The sample size is also impacted by the kind of sampling technique selected by the researcher for conducting the research. For example, when a researcher adopts the simple random sampling technique then he must choose large sample size, but on the other hand in stratified sampling technique the researcher needs a comparatively smaller sample size for accurate and effective results.

7) Attitude of Respondents : The researcher is an experienced individual, so he will always choose a larger sample size in which he thinks that the non-response rate will be higher. In simple words, if there are chances that the respondents will not reply and return the questionnaire, then large sample size is chosen.

8) Degree of the Variability : Degree of variability specifies the extent to which the scores in any distribution are spread in the population. Larger numbers always indicate greater variability of scores in a population. Sometimes dispersion is substituted for variability, and this term is widely used in statistics. If the population is heterogeneous, then the researcher uses larger sample size so as to receive a specified precision level. On the other hand, if the population is more homogeneous then the smaller sample size is selected by the researcher.

Types/ Methods of Sampling



Probability Sampling:

Probability sampling refers to the selection of a sample from a population, when this selection is based on the principle of randomization, that is, random selection or chance. Probability sampling is more complex, more time-consuming and usually more costly than non-probability sampling.

Simple random sampling

Simple random sampling is a statistical method in which everyone in a population has an equal chance of being selected into a sample. The sample represents a smaller and more manageable portion of the people that can be studied and analyzed. It's a fundamental technique to gather data and make inferences about a population.

Simple random sampling is considered a fair and unbiased sample selection method. This type of sampling is the most straightforward sample selection method.

What is Simple Random Sampling?

Simple random sampling is a technique where every item in the population has an even chance and likelihood of being selected. Here, the selection of items entirely depends on luck or probability. Therefore, this sampling technique is also a method of chance.

Simple random sampling is a fundamental method and can easily be a component of a more complex method. The main attribute of this sampling method is that every sample has the same probability of being chosen.

The sample size in simple random sampling method should ideally be more than a few hundred so that it can be applied appropriately. This method is theoretically simple to understand but difficult to implement practically. Working with a large sample size isn't an easy task, and it can sometimes be challenging to find a realistic sampling bias frame.

Simple Random Sampling Methods

Researchers follow these methods to select a simple random sample:

1. They prepare a list of all the population members initially, and each member is marked with a specific number (for example, if there are n members, then they will be numbered from 1 to N).
2. Researchers from this population choose random samples using random number tables and random number generator software. Researchers prefer random number generator software, as no human interference is necessary to generate samples.

Two approaches aim to minimize any biases in the process of this method:

- **Method of lottery**

Using the lottery method is one of the oldest ways and is a mechanical example of random sample. Researchers draw numbers from the box randomly to choose samples. In this method, the researcher gives each member of the population a number.

- **Use of random numbers**

Using random numbers is an alternative method that also involves numbering the population. A numbered table similar to the one below can help with this sampling technique.

Using random numbers is an alternative method that also involves numbering the population. A numbered table similar to the one below can help with this sampling technique.

Random Number Table									
1	69	24	40	68	29	39	95	60	30
97	23	70	59	79	4	47	19	38	20
13	44	5	71	12	99	78	34	9	96
34	55	83	21	72	3	37	85	61	2
22	80	18	82	54	32	84	16	46	88
7	43	6	48	11	92	63	53	86	28
56	90	36	91	64	45	15	73	10	87
49	65	50	14	51	33	89	52	74	57
98	17	100	58	5	8	77	25	62	31
27	76	66	81	26	93	41	94	67	42

Simple Random Sampling Formula

Consider that a hospital has 1000 staff members and must allocate a night shift to 100 members. All their names will be put in a bucket to be randomly selected. Since each person has an equal chance of being selected. Since we know the population size (N) and sample size (n), the calculation can be as follows:

$$P=1- \frac{N-1}{N} \cdot \frac{N-2}{N-1} \dots \frac{N-n}{N-(n-1)}$$

$$\text{Cancelling} = 1 - \frac{N-n}{N}$$

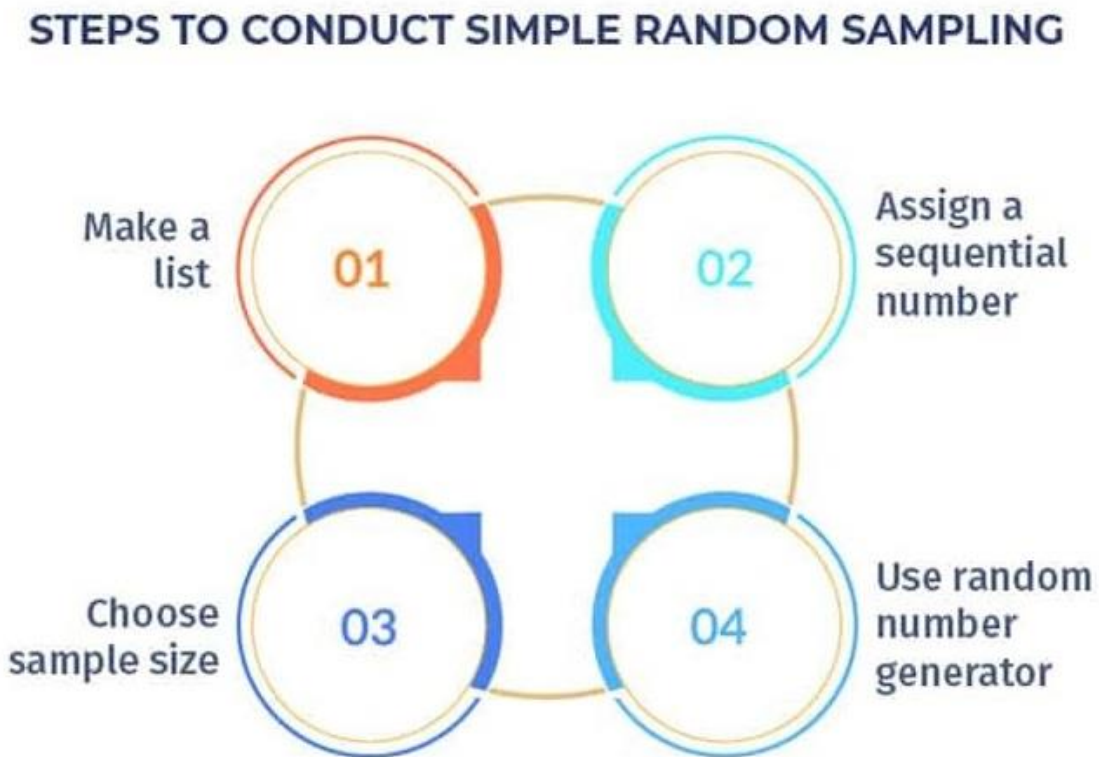
$$= \frac{n}{N}$$

$$= \frac{100}{1000}$$

$$= 10\%$$

Simple Random Sampling Steps

Follow these steps to extract a simple random sample of 100 employees out of 500.



1. **Make a list** of all the employees working in the organization. (as mentioned above, there are 500 employees in the organization, so the record must contain 500 names).
2. **Assign a sequential number** to each employee (1,2,3...n). This is your sampling frame (the list from which you draw your sample).
3. **Figure out what your sample size is going to be.** (In this case, the sample size is 100).
4. **Use a random number generator** to select the sample, using your frame (population size) from Step 2 and your sample size from Step 3. For example, if your sample size is 100 and your population is 500, generate 100 random numbers between 1 and 500.

It is important to note that Simple Random Sampling is just one of many sampling methods available, and it may not always be the best option for your specific research needs.

Simple Random Sampling in Research

Today's market research projects are much larger and involve an indefinite number of items. It is practically impossible to study every member of the population's thought process and derive inference from the study.

If, as a researcher, you want to save your time and money, simple random sampling is one of the best probability sampling methods that you can use. Getting data from a sample is more advisable and practical.

Using a census or a sample depends on several factors, such as the type of census, the degree of homogeneity/heterogeneity, costs, time, feasibility to study, the degree of accuracy needed, etc.

Advantages of Simple Random Sampling

Simple random sampling has several advantages, including:

1. It is a fair sampling method, and if applied appropriately, it helps reduce any bias involved compared to any other sampling method.
2. Since it involves a large sample frame, it is usually easy to pick a smaller sample size from the existing larger population.
3. The person conducting the research doesn't need to have prior knowledge of the data he/she is collecting. One can ask a question to gather the researcher need not be a subject expert.
4. This sampling method is a fundamental method of collecting the data. You don't need any technical knowledge. You only require essential listening and recording skills.
5. Since the population size is vast in this type of sampling method, there is no restriction on the sample size that the researcher needs to create. From a larger population, you can get a small sample quite quickly.

6. The data collected through this sampling method is well informative; the more samples better is the quality of the data.

Overall, this is a valuable and versatile method for gathering data and making inferences about populations.

Systematic sampling

Systematic sampling is a statistical method used to select a sample from a larger population systematically and randomly. It is a widespread technique for researchers and analysts who want to gather data from a large population without surveying every individual. This method is beneficial when the population is large, diverse, or hard to reach.

This article will delve into the details of systematic sampling, including its definition and advantages. We will also provide examples of how it can be used in various research and data analysis contexts and tips on implementing it effectively. Whether you are a researcher, analyst, or simply interested in understanding this sampling method, this article will give you an understanding of what it is and how to use it for better decision-making.

What is systematic sampling?

Systematic sampling is a statistical method that researchers use to zero down on the desired population they want to research. Researchers calculate the sampling interval by dividing the entire population size by the desired sample size determination. It is an extended implementation of probability sampling in which each group member is selected regularly to form a sample.

Systematic sampling definition

Systematic sampling is a probability sampling method where the researcher chooses elements from a target population by selecting a random starting point and selecting sample members after a fixed 'sampling interval.'

For example, in school, while selecting the captain of a sports team, most of our coaches asked us to call out numbers such as 1-5 (1-n) and the students with a random number decided by the coach. For instance, three would be called out as team captains. It is a non-stressful selection process for both the coach and the players. There's an equal opportunity for every member of a population to be selected using this sampling bias technique.

What are the steps to form a sample using the systematic sampling technique?

Here are the steps to form a systematic sample:

Step one: Develop a defined structural audience to start working on the sampling aspect.

Step two: As a researcher, figure out the ideal size of the sample, i.e., how many people from the entire population to choose to be a part of the sample.

Step three: Once you decide the sample size, assign a number to every member of the sample.

Step four: Define the interval of this sample. This will be the standard distance between the elements.

For example, the sample interval should be 10, which is the result of the division of 5000 (N= size of the population) and 500 (n=size of the sample).

$$\text{Systematic Sampling Formula for interval (i)} = N/n = 5000/500 = 10$$

Step five: Select the members who fit the criteria which in this case will be 1 in 10 individuals.

Step six: Randomly choose the starting member (r) of the sample and add the interval to the random number to keep adding members in the sample. r, r+i, r+2i, etc. will be the elements of the sample.

How systematic sampling works

When you are sampling, ensure you represent the population fairly. Systematic sampling is a symmetrical process where the researcher chooses the samples after a specifically defined interval. Sampling like this leaves the researcher no room for bias regarding choosing the sample. To understand how it exactly works, take the example of the gym class where the instructor asks the students to line up and asks every third person to step out of the line. Here, the instructor has no influence over choosing the samples and can accurately represent the class.

Systematic sampling example

For instance, if a local NGO is seeking to form a systematic sample of 500 volunteers from a population of 5000, they can select every 10th person in the population to build a sample systematically.

Systematic Sampling Types

Here are the types of systematic sampling:

1. Systematic random sampling
2. Linear systematic sampling
3. Circular systematic sampling

Let's take a closer look at these sampling techniques.

Systematic random sampling:

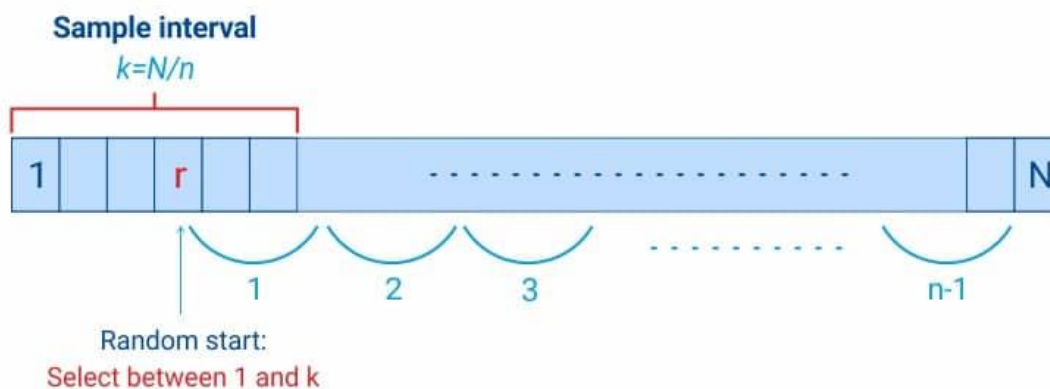
Systematic random sampling is a method to select samples at a particular preset interval. As a researcher, select a random starting point between 1 and the sampling interval. Below are the example steps to set up a systematic random sample:

1. First, calculate and fix the sampling interval. (The number of elements in the population divided by the number of elements needed for the sample.)
2. Choose a random starting point between 1 and the sampling interval.
3. Lastly, repeat the sampling interval to choose subsequent elements.

Linear systematic sampling:

Linear systematic sampling is a method where samples aren't repeated at the end and 'n' units are selected to be a part of a sample having 'N' population units. Rather than selecting these 'n' units of a sample randomly, a researcher can apply a skip logic to select these. It follows a linear path and then stops at the end of a particular population.

This sampling or skip interval (k) = N (total population units)/ n (sample size)

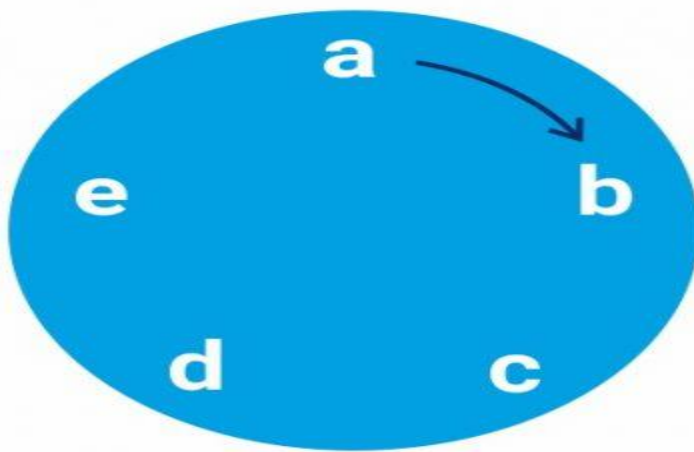


How is a Linear systematic sample selected?

- Arrange the entire population in a classified sequence.
- Select the sample size (n)
- Calculate sampling interval (k) = N/n
- Select a random number between 1 to k (including k)
- Add the sampling interval (k) to the chosen random number to add the next member to a sample and repeat this procedure to add remaining members of the sample.
- In case k isn't an integer, you can select the closest integer to N/n .

Circular systematic sampling:

In circular systematic sampling, a sample starts again from the same point once again after ending; thus, the name. For example, if $N = 7$ and $n = 2$, $k=3.5$. There are two probable ways to form sample:



1. If we consider $k=3$, the samples will be – ad, be, ca, db and ec.
2. If we consider $k=4$, the samples will be – ae, ba, cb, dc and ed.

How is a circular systematic sample selected?

- Calculate sampling interval (k) = N/n . (If $N = 11$ and $n = 2$, then k is taken as 5 and not 6)
- Start randomly between 1 to N
- Create samples by skipping through k units every time until you select members of the entire population.
- In the case of this method, there will be N number of samples, unlike k samples in the linear systematic sampling method.

Difference between linear systematic sampling and circular systematic sampling:

Here is the difference between linear and circular.

Linear	Circular
Create samples = k (sampling interval)	Create samples = N (total population)
The start and endpoints of this sample are distinct.	It restarts from the start point once the entire population is considered.
All sample units should be arranged in a linear manner prior to selection.	Elements will be arranged in a circular manner.

What are the advantages of systematic sampling?

Here are the advantages:

- It's extremely simple and convenient for the researchers to create, conduct, analyze samples.
- As there's no need to number each member of a sample, it is better for representing a population in a faster and simpler manner.
- The samples created are based on precision in member selection and free from favoritism.
- In the other methods of probability sampling methods such as cluster sampling and stratified sampling or non-probability methods such as convenience sampling, there are chances of the clusters created to be highly biased which is avoided in systematic sampling as the members are at a fixed distance from one another.
- The factor of risk involved in this sampling method is extremely minimal.
- In case there are diverse members of a population, this sampling technique can be beneficial because of the even distribution of members to form a sample.

Other probability sampling techniques like cluster sampling and stratified random sampling can be very unorganized and challenging due to which researchers and statisticians have turned to methods like systematic sampling or simple random sampling for better sampling results. It consumes the least time as it requires a selection of sample size and identification of the starting point for this sample, which needs to be continued at regular intervals to form a sample.

When to use systematic sampling?

Let's take an example where you want to form a sample of 500 individuals out of a population of 5000; you'd have to number every person in the population.

Once the numbering is done, the researcher can select a number randomly, for instance, 5. The 5th individual will be the first to be a part of the systematic sample. After that, the 10th member will be added into the sample, so on and so forth (15th, 25th, 35, 45th, and members till 4995).

Here are 4 other situations of when to use Systematic Sampling:

1. **Budget restrictions:** In comparison to other sampling methods like simple random sampling, this sampling technique is more suitable for conditions where there are budget restrictions and also the extremely uncomplicated accomplishment of the study.
2. **Uncomplicated implementation:** As systematic sampling depends on the defined sampling intervals to decide the sample, it becomes simple for the researchers and statisticians to manage samples with more respondents. This is because the time invested in creating samples is minimal, and the cost spent is also restricted due to the periodic nature of systematic sampling.
3. **Absence of data pattern:** There are specific data that don't have an arrangement in place. This data can be analyzed in an unbiased manner, using systematic sampling.
4. **Low risk of data manipulation in research:** It is highly productive while researching a broad subject, especially when there's a negligible risk of data manipulation.

Stratified random sampling

Stratified random sampling is a widely used statistical technique in which a population is divided into different subgroups, or strata, based on some shared characteristics. The purpose of stratification is to ensure that each stratum in the sample and to make inferences about specific population subgroups.

This technique is beneficial in cases where the population is heterogeneous, and a simple random sample may not provide accurate results. By dividing the population into strata, researchers can ensure that their sample is representative of the population and avoid sampling biases.

What is Stratified Random Sampling?

Stratified random sampling is a type of probability method using which a research organization can branch off the entire population into multiple non-overlapping, homogeneous groups (strata) and randomly choose final members from the various strata for research which reduces cost and improves efficiency. Members in each of these groups should be distinct so that every member of all groups gets an equal opportunity to be selected using simple probability. This sampling method is also called "random quota sampling."

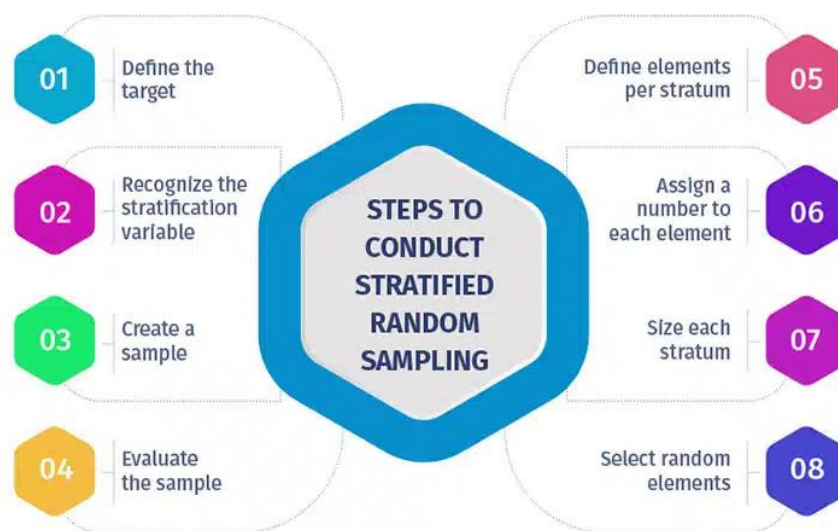
Let's consider a situation where a research team seeks opinions about religion among various age groups. Instead of collecting feedback from 326,044,985 U.S citizens, random samples of around 10000 can be selected for research. These 10000 citizens can be divided into groups according to age, i.e., 18-29, 30-39, 40-49, 50-59, and 60 and above. Each stratum will have distinct members

and the number of members—age, socioeconomic divisions, nationality, religion, educational achievements, and other classifications.

8 Steps to Conduct Stratified Random Sampling

The following are the steps to select a stratified random sample:

The following are the steps to select a stratified random sample:



1. Define the target audience.
2. Recognize the stratification variable or variables and figure out the number of strata to be used. These stratification variables should be in line with the objective of the research. Every additional information decides the stratification variables. For instance, if the objective of the research is to understand all the subgroups, the variables will be related to the subgroups.
3. Use an already-existent sampling frame or create a frame that's inclusive of all the information of the stratification variable for all the elements in the target audience.
4. Make changes after evaluating the sampling frame on the basis of lack of coverage, over-coverage, or grouping.
5. Considering the entire population, each stratum should be unique and should cover each and every member of the population. Within the stratum, the differences should be minimum, whereas each stratum should be extremely different from one another. Each element of the population should belong to just one stratum.
6. Assign a random, unique number to each element.

7. Figure out the size of each stratum according to your requirement. The numerical distribution amongst all the elements in all the strata will determine the type of sampling to be implemented. It can either be proportional or disproportional stratified sampling.
8. The researcher can select random elements from each stratum to form the sample. A minimum of one piece must be chosen from each stratum so that there's representation from every stratum, but if two elements from each stratum are selected, quickly calculate the error margins of the calculation of collected data.

Using this method helps ensure that the sample is representative of the population and reduces sampling error, leading to more accurate results.

Types of Stratified Random Sampling

- **Proportionate Sampling:**

In this approach, each stratum sample size is directly proportional to the population size of the entire population of strata. That means each strata sample has the same sampling fraction.

Proportionate Stratified Random Sampling Formula: $n_h = (N_h / N) * n$

n_h = Sample size for h^{th} stratum

N_h = Population size for h^{th} stratum

N = Size of entire population

n = Size of entire sample

Stratum	A	B	C	D
Population Size	500	1000	1500	2000
Sampling Fraction	1/2	1/2	1/2	1/2
Final Sampling Size Results	250	500	750	1000

If you have four strata with 500, 1000, 1500, and 2000 respective sizes, the research organization selects $\frac{1}{2}$ as the sampling fraction. A researcher must choose 250, 500, 750, and 1000 members from the separate stratum.

Irrespective of the sample size of the population, the sampling fraction will remain uniform across all the strata.

- **Disproportionate Sampling:**

The sampling fraction is the primary differentiating factor between proportionate and disproportionate stratified random sampling. In excessive sampling, each stratum will have a different sampling fraction.

The success of this sampling method depends on the researcher's precision at fraction allocation. If the allotted fractions aren't accurate, the results may be biased due to the overrepresented or underrepresented strata.

Stratum	A	B	C	D
Population Size	500	1000	1500	2000
Sampling Fraction	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$
Final Sampling Size Results	250	333	375	400

Stratified Random Sampling Examples

Researchers and statisticians use stratified random sampling to analyze relationships between two or more strata. As stratified random sampling involves multiple layers or strata, it's crucial to calculate the strata before calculating the sample value.

Following is a classic stratified random sampling example:

Let's say 100 (N_h) students of a school having 1000 (N) students were asked questions about their favorite subject. It's a fact that the students of the 8th grade will have different subject preferences

than the students of the 9th grade. For the survey to deliver precise results, the ideal manner is to divide each step into various strata.

Here's a table of the number of students in each grade:

Grade	Number of students (n)	Calculate the sample of each grade using the formula:
5	150	Stratified Sample (n_5) = $100 / 1000 * 150 = 15$
6	250	Stratified Sample (n_6) = $100 / 1000 * 250 = 25$
7	300	Stratified Sample (n_7) = $100 / 1000 * 300 = 30$
8	200	Stratified Sample (n_8) = $100 / 1000 * 200 = 20$
9	100	Stratified Sample (n_9) = $100 / 1000 * 100 = 10$

Advantages of Stratified Random Sampling

Stratified random sampling is a statistical technique that offers several advantages over other methods. Some of the critical benefits include the following:

- Better accuracy in results in comparison to other probability sampling methods such as cluster sampling, simple random sampling, and systematic sampling or non-probability methods such as convenience sampling. This accuracy will be dependent on the distinction of various strata.
- Convenient to train a team to stratify a sample due to the exactness of the nature of this sampling technique.
- Due to statistical accuracy of this method, smaller sample sizes can also retrieve highly useful results for a researcher.
- This sampling technique covers maximum population as the researchers have complete charge over the strata division.
- Health surveys can benefit from this method to represent the population concerning health status, disease prevalence, or risk factors.

Stratified random sampling is a valuable and efficient method for a population with distinct subgroups. Its ability to increase precision, represent each subgroup proportionately, control for bias, save costs, and improve data quality make it a valuable tool for researchers in many fields.

When to use Stratified Random Sampling?

Stratified random sampling should be used when the population under study has distinct subgroups, and the goal is to make inferences about each subgroup or the population as a whole. This technique is beneficial when:

1. Stratified random sampling is a highly productive method sampling in situations where the researcher intends to focus only on specific strata from the available population data. This way, the desired characteristics of the strata can be found in the survey sample.
2. Researchers rely on this sampling method in cases where they intend to establish a relationship between two or more different strata. If this comparison is conducted using simple random sampling, there is a higher likelihood of the target groups not being equally represented.
3. Samples with a population which are difficult to access or contact can be quickly involved in the research process using the stratified random sampling technique.
4. The accuracy of statistical results is higher than simple random sampling since the elements of the sample are chosen from relevant strata. The diversification within the strata will be much lesser than the diversification in the target population. Due to the accuracy involved, the required sample size will be much lesser, which will help researchers save time and effort.
5. Public opinion polls often use stratified random sampling to represent the population concerning demographic characteristics such as region, political affiliation, or age.

Stratified random sampling is practical when the population has distinct subgroups, and the goal is to make inferences about each subgroup or the population as a whole.

Cluster sampling

Cluster sampling is defined as a sampling method where the researcher creates multiple clusters of people from a population where they are indicative of homogeneous characteristics and have an equal chance of being a part of the sample.

Consider a scenario where a data organization is looking to survey the performance of smartphones across Germany. They can divide the entire country's population into cities (clusters), select further towns with the highest population, and filter those using mobile devices.

What is Cluster Sampling?

Cluster sampling is a probability sampling technique where researchers divide the population into multiple groups (clusters) for research. So researchers then select random groups with a simple random or systematic random sampling technique for data collection and unit of analysis.

Example: A researcher wants to conduct a study to judge the performance of sophomores in business education across the U.S. It is impossible to conduct a research study that involves a student in every university. Instead, cluster sampling allows the researcher to club the universities from each city into one cluster. These clusters then define the sophomore student population in the U.S. Next, either using simple random sampling or systematic random sampling and randomly picking clusters for the research study. Subsequently, by using simple or systematic sampling, the sophomore's from each of these selected clusters can be chosen on whom to conduct the research study.

In this sampling technique, researchers analyze a sample that consists of multiple sample parameters such as demographics, habits, background – or any other population attribute, which may be the focus of conducted research. This method is usually conducted when groups that are similar yet internally diverse form a statistical population. Instead of selecting the entire population, cluster sampling allows the researchers to collect data by bifurcating the data into small, more productive groups.

Types of Cluster Sampling

There are two ways to classify this sampling technique. The first way is based on the number of stages followed to obtain the cluster sample, and the second way is the representation of the groups in the entire cluster analysis. In most cases, sampling by clusters happens over multiple stages. A stage is considered to be the step taken to get to the desired sample. We can divide this technique into single-stage, two-stage, and multiple stages.

Single-stage Cluster Sampling:

As the name suggests, sampling is done just once. An example of single-stage cluster sampling – An NGO wants to create a sample of girls across five neighboring towns to provide education. Using single-stage sampling, the NGO randomly selects towns (clusters) to form a sample and extend help to the girls deprived of education in those towns.

Two-stage Cluster Sampling:

Here, instead of selecting all the elements of a cluster, only a handful of members are chosen from each group by implementing systematic or simple random sampling. An example of two-stage cluster sampling – A business owner wants to explore the performance of his/her plants that are spread across various parts of the U.S. The owner creates clusters of plants. So he/she then select random samples from these clusters to conduct research.

Multiple-stage Cluster Sampling:

Multiple-stage cluster sampling takes a step or a few steps further than two-stage sampling.

For conducting effective research across multiple geographies, one needs to form complicated clusters that can be achieved only using the multiple-stage sampling technique. An example of Multiple stage sampling by clusters – An organization intends to survey to analyze the performance of smartphones across Germany. They can divide the entire country's population into cities (clusters) and select cities with the highest population and also filter those using mobile devices.

Steps to conduct Cluster Sampling

Here are the steps to perform cluster sampling:



1. **Sample:** Decide the target audience and also the sample size.
2. **Create and evaluate sampling frames:** Create a sampling frame by using either an existing framework or creating a new one for the target audience. Evaluate frameworks based on coverage and clustering and make adjustments accordingly. These groups will be varied, considering the population, which can be exclusive and comprehensive. Members of a sample are selected individually.
3. **Determine groups:** Determine the number of groups by including the same average members in each group. So make sure each of these groups is distinct from one another.
4. **Select clusters:** Choose clusters by applying a random selection.

5. **Create sub-types:** It is bifurcated into two-stage and multi-stage subtypes based on the number of steps followed by researchers to form clusters.

Applications of Cluster Sampling

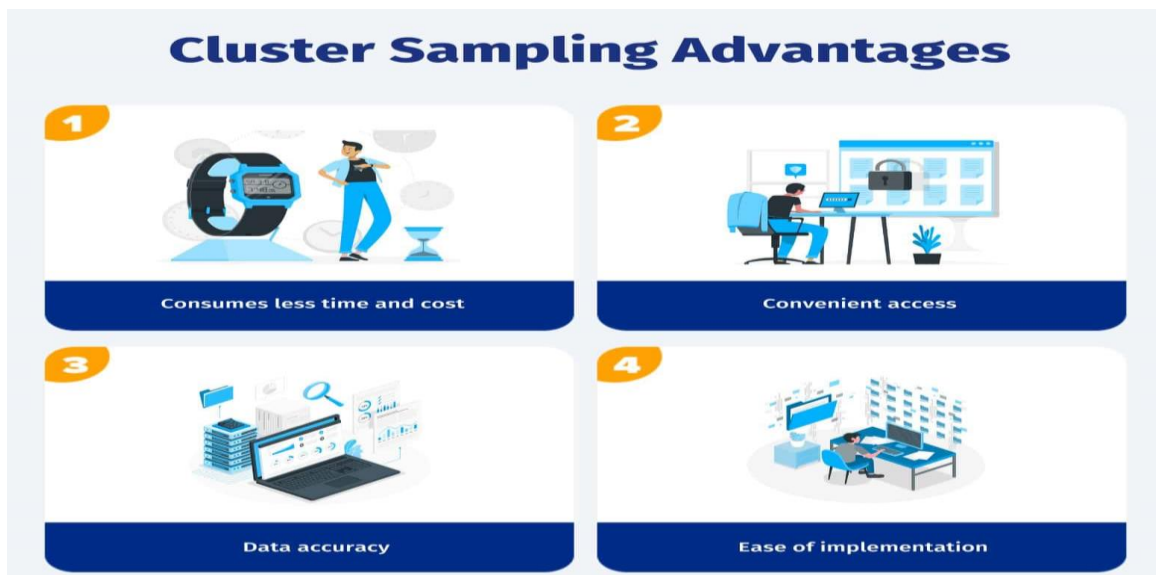
This sampling technique is used in an area or geographical cluster sampling for market research. A broad geographic location can be expensive to survey compared to surveys sent to clusters divided based on region. The sample numbers must be increased to achieve accurate results, but the cost savings make this process of rising clusters attainable.

Cluster Sampling in statistics

The technique is widely used in statistics where the researcher can't collect data from the entire population. So, it is the most economical and practical solution for research statisticians. Take the example of a researcher looking to understand smartphone usage in Germany. In this case, the cities of Germany will form clusters. This sampling method is also used in wars and natural calamities to draw inferences about a population, where collecting data from every individual residing is impossible.

Cluster Sampling Advantages

There are multiple advantages to using cluster sampling. Here they are:



- **Consumes less time and cost:** Sampling of geographic segmentation divided groups requires less work, time, and cost. It's a highly economical method to observe clusters instead of randomly doing them throughout a particular region by allocating a limited number of resources to those selected clusters.
- **Convenient access:** Researchers can choose large samples with this convenience sampling technique, and that'll increase accessibility to various clusters.
- **Data accuracy:** Since there can be large samples in each cluster, loss of data accuracy in information per individual can be compensated.
- **Ease of implementation:** Cluster sampling facilitates information from various areas and groups. Researchers can quickly implement it in practical situations compared to other probability sampling methods.

In comparison to simple random sampling, this technique can be useful in deciding the characteristics of a group such as population, and researchers can implement it without having a sampling frame for all the elements for the entire population.

Cluster Sampling vs Stratified Sampling

Since cluster sampling and stratified random sampling are pretty similar, there could be issues with understanding their finer nuances. So let's talk about the significant differences between cluster sampling and stratified sampling:

Cluster Sampling	Stratified Sampling
Elements of a population are randomly selected to be a part of groups (clusters).	The researcher divides the entire population into even segments (strata).
Members from randomly selected clusters are a part of this sample.	Researchers consider individual components of the strata randomly to be a part of sampling units.
Researchers maintain homogeneity between clusters.	Researchers maintain homogeneity within the strata.
Researchers divide the clusters naturally.	The researchers or statisticians primarily decide the strata division.
The key objective is to minimize the cost involved and enhance competence.	The key objective is to conduct accurate sampling, along with a properly represented population.

Non Probability Sampling:

Non-probability sampling is a method of selecting units from a population using a subjective (i.e. non-random) method. Since non-probability sampling does not require a complete survey frame, it is a fast, easy and inexpensive way of obtaining data.

Convenience sampling

Convenience sampling is a simple and easy way to get information compared to other sampling methods. Most of the time, simple and easy go well together. But you need to know what it is so you know when to use it and when not to.

It is a type of sampling that doesn't depend on chance and is often used in research studies. This sampling technique involves choosing people who are easy for the researcher to reach and get in touch with.

Instead of picking people at random from a certain population, convenience sampling involves picking the people who are easiest for the researcher to get information from.

Convenience sampling is often used when other types of sampling methods are hard or impossible to use because of time, cost, or other issues. Even though it can be a quick and easy way to get data, it can also have biases and limitations that can affect how well the results can be used in the real world and how reliable they are.

What is convenience sampling?

Convenience sampling is defined as a method adopted by researchers where they collect market research data from a conveniently available pool of respondents.

It is the most commonly used sampling as it's incredibly prompt, uncomplicated, and economical. Members are often readily approachable to be a part of the sample.

Researchers use various sampling techniques in situations where there are large populations. In most cases, testing the entire community is practically impossible because they are not easy to reach. Researchers use convenience sampling in situations where additional inputs are not necessary for the principal research. There are no criteria required to be a part of this sample.

Thus, it becomes incredibly simplified to include elements in this sample. All components of the population are eligible and dependent on the researcher's proximity to get involved in the sample.

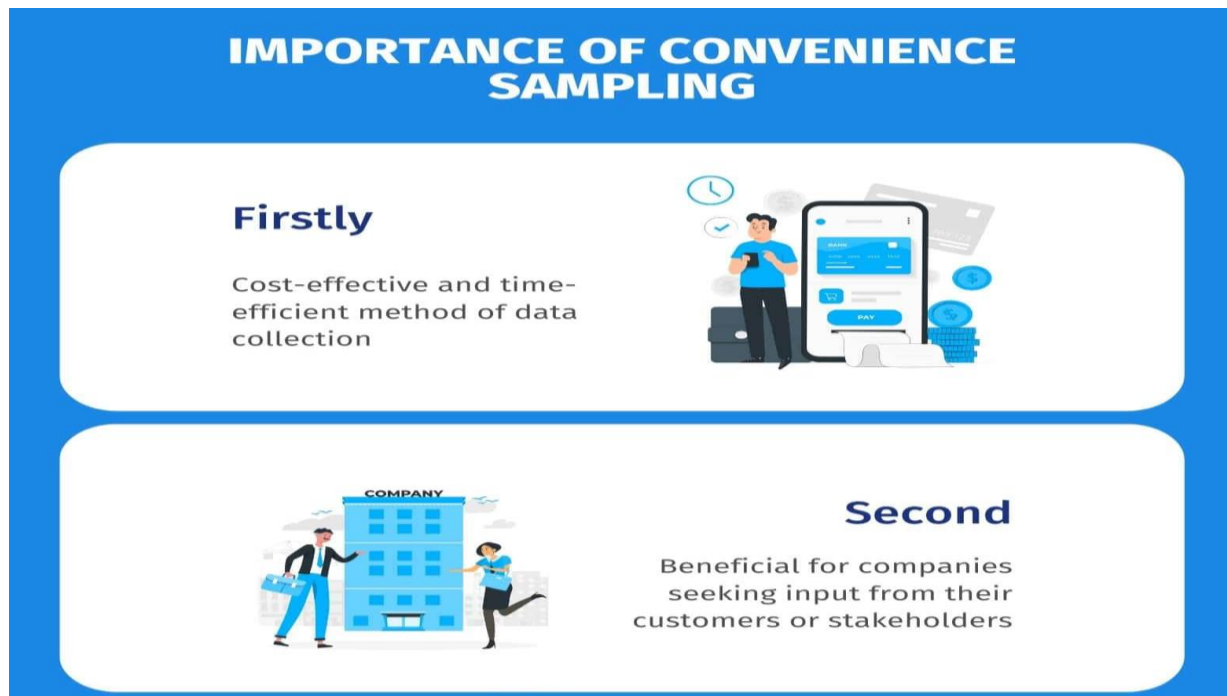
The researcher chooses members merely based on proximity and doesn't consider whether they represent the entire population or not.

Using this technique, they can observe habits, opinions, and viewpoints in the easiest possible manner.

A good example of convenience sampling is A new NGO that wants to establish itself in 20 cities. It selects the top 20 cities to serve based on the proximity to where they're based.

Importance of convenience sampling

Convenience sampling is beneficial to businesses in a variety of ways.



- Firstly, it can be a cost-effective and time-efficient method of data collection. Businesses can save time and money by picking readily available participants instead of employing more elaborate sampling procedures.
- Secondly, it can be beneficial for companies seeking input from their customers or stakeholders. Businesses can gain useful insights and feedback by selecting individuals who have interacted with the firm or used its products/services.

It is important to highlight that convenience sampling has limits and biases that can affect the reliability and generalizability of the findings.

As a result, before depending on this sampling technique to make crucial decisions, organizations must carefully analyze the potential biases and limits.

Applications of convenience sampling:

Convenience sampling is applied by brands and organizations to measure their perception of their image in the market. Data is collected from potential customers to understand specific issues or manage opinions of a newly launched product. In some cases, it is the only available option.

For example, a university student working on a project and wants to understand the average consumption of soda on campus on a Friday night will most possibly call his/her classmates and friends and ask how many cans of soda they consume. Or may go to a party nearby and conduct an easy online survey.

There is always a chance that the randomly selected population may not accurately represent the population of interest, thus increasing the chances of research bias.

Example of convenience sampling

A basic example of a convenience sampling method is when companies distribute their promotional pamphlets and ask questions at a mall or on a crowded street with randomly selected participants.

Businesses use this sampling technique to gather information to address critical issues arising from the market. They also use it when collecting feedback about a particular feature or a newly launched product from the sample created.

During the initial stages of survey research, researchers usually prefer using this as it's quick and easy to deliver results. Even if many statisticians avoid implementing this technique, it is vital in situations where you intend to get insights in a shorter period or without investing too much money.

For instance, a marketing student needs to get feedback on the “scope of content marketing in 2020.” The student may quickly create an online survey, send a link to all the contacts on your phone, share a link on social media, and talk to people you meet daily, face-to-face.

What is convenience sampling in qualitative research?

Convenience sampling is a qualitative research sampling strategy that involves selecting participants based on their accessibility and availability to the researcher. Rather than being drawn at random from a bigger population, participants in this strategy are picked because they are easily available to the researcher.

It is frequently used in qualitative research because it can be a cost-effective and time-efficient way to collect data from individuals who are difficult to reach or may not be interested in participating in the study.

For example, a researcher conducting a study on cancer survivors' experiences may choose to recruit volunteers using social media groups or cancer support networks because they are easily accessible and likely to be interested in the study.

Convenience sampling in qualitative research can also contribute to data biases because the sample may not be representative of the greater community or may be prejudiced towards participants with certain qualities or experiences.

While evaluating the findings of their study, researchers must carefully evaluate the various limits and biases of this sampling strategy.

Advantages of using convenience sampling

Here are the advantages of adopting a convenience sampling approach:



1. Collect data quickly

In situations where time is a constraint, many researchers choose this method for quick data collection. The rules to gather elements for the sample are the least complicated in comparison to techniques such as simple random sampling, stratified sampling, and systematic sampling. Due to this simplicity, data collection takes minimal time.

2. Inexpensive to create samples
The money and time invested in other probability sampling methods are quite large compared to convenience sampling. It allows researchers to generate more samples with less or no investment and in a brief period.
3. Easy to do research
The name of this surveying technique clarifies how samples are formed. Elements are easily accessible by the researchers and so, collecting members for the sample becomes easy.
4. Low cost
Low cost is one of the main reasons why researchers adopt this technique. When on a small budget, researchers – especially students, can use the budget in other areas of the project.
5. Readily available sample
Data collection is easy and accessible. Most convenience sampling considers the population at hand. Samples are readily available to the researcher. They do not have to move around too much for data collection. Quotas are met quickly, and the data collection can commence even within a few hours.
6. Fewer rules to follow
It doesn't require going through a checklist to filter members of an audience. Here, gathering critical information and data becomes uncomplicated. For instance, if an NGO wants to survey women's empowerment, they can go to schools, colleges, offices, etc., in their proximity and gather quick responses.

Disadvantages of convenience sampling

Convenience sampling includes a number of drawbacks that can reduce the reliability and validity of research findings. The following are some of the major drawbacks:

1. Bias in sampling
When participants are not chosen at random from a larger population, this might result in sampling bias. This indicates that the sample may not be typical of the greater population, and the findings may not apply to other groups.
2. Lack of variety
Because the researcher may choose participants who are easily accessible and have comparable features, this may result in a lack of variety in the sample. This can potentially narrow the spectrum of opinions and experiences represented in the sample.
3. External validity is limited
It may have poor external validity due to the possibility of sampling bias and a lack of diversity. As a result, the findings may not apply to different locations or populations.
4. Unknown errors
Convenience sampling can lead to unknown errors since the researcher may be unaware of how skewed or unrepresentative the sample is of the population.
5. Possibility of researcher bias
Researchers may be more prone to select participants who they believe would offer data that supports their theory, perhaps contributing to researcher bias.

While it can be a valuable and cost-effective method of data collection in some contexts, it is crucial to understand its limitations and inherent biases. When selecting to utilize it in their research, researchers should carefully assess its advantages and disadvantages.

How to reduce bias in convenience sampling?

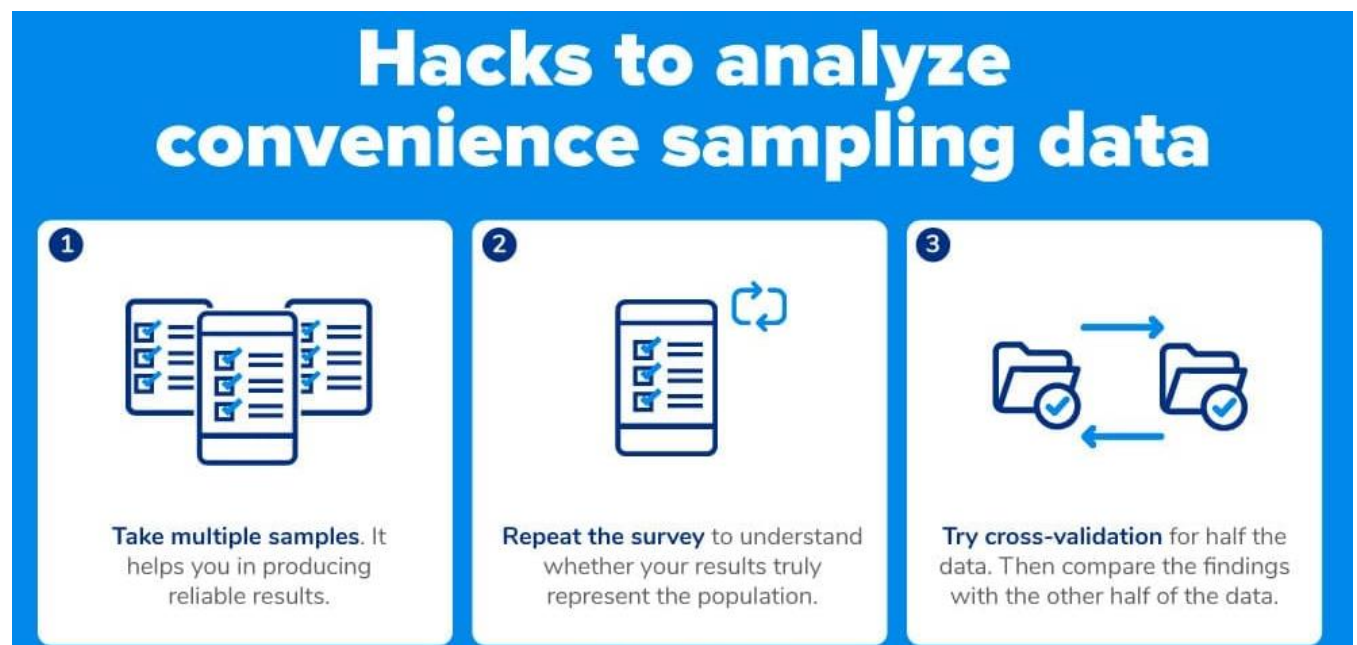
The best way of reducing bias in convenience sampling is to use it along with probability sampling. Since it is usually biased, probability sampling gets the measurement parameter with it to keep this sampling bias under check.

After receiving a fair idea about this bias using probability sampling, the researcher can use both convenience sampling and probability sampling techniques to draw a more accurate estimation. The probability aspect used, along with the convenience sample, will have to be powerful enough to overcome it.

Bias can make the entire sample futile, which is the last thing a researcher needs. This bias can be reduced or eliminated by including probability sampling.

How to efficiently analyze convenience sampling data?

Here are three quick hacks to efficiently analyze convenience sampling data. It is best to use probability sampling, but when that is not possible, here are three hacks you should keep in mind.



- Take multiple samples. It helps you in producing reliable results.
- Repeat the survey to understand whether your results truly represent the population.

- For a big sample size, try cross-validation for half the data. Then compare the findings with the other half of the data.

Judgmental sampling

Definition

Judgmental sampling, also called purposive sampling or authoritative sampling, is a non-probability sampling technique in which the sample members are chosen only on the basis of the researcher's knowledge and judgment. As the researcher's knowledge is instrumental in creating a sample in this sampling technique, there are chances that the results obtained will be highly accurate with a minimum margin of error.

The process of selecting a sample using judgmental sampling involves the researchers carefully picking and choosing each individual to be a part of the sample. The researcher's knowledge is primary in this sampling process as the members of the sample are not randomly chosen.



When to execute Judgmental Sampling?

Judgmental sampling is most effective in situations where there are only a restricted number of people in a population who own qualities that a researcher expects from the target population. Researchers prefer to implement Judgmental sampling when they feel that other sampling techniques will consume more time and that they have confidence in their knowledge to select a sample for conducting research.

Judgmental or Expert sampling is usually used in situations where the target population comprises of highly intellectual individuals who cannot be chosen by using any other probability or non-

probability sampling technique. It is also used in situations where the sample selected using other sampling methods need to be approved or filtered. For instance, in situations where a researcher conducts convenience sampling to gather feedback from professors about their university but the fact that there are high chances of the results to be skewed, researchers prefer judgmental sampling to select those professors who will provide 100% feedback about the university.

Selecting each individual of the sample is a critical challenge that an intellectual researcher will undertake. It is a tedious task to handpick members of a sample while ensuring there is no bias involved.

The authority involved in the selection process may not necessarily be “experts” in the field but they have to comply with certain characteristics expected from a Judgmental sampling authority and sampling errors. Education or work experience is not considered while appointing authorities for the selection process.

Purposive sampling is used where there is time-constraint for sample creation and the authorities involved would prefer relying on their knowledge and not on other sampling methods. But, one must keep in mind, the fact that a researcher may or may not have the appropriate proficiency to conduct an effective sampling process. This is the only disadvantage of purposive sampling. Each researcher who takes up the responsibility of creating a sample using expert sampling will have to be extremely confident in their own skills and understanding of the subject.

Examples of Judgmental Sampling

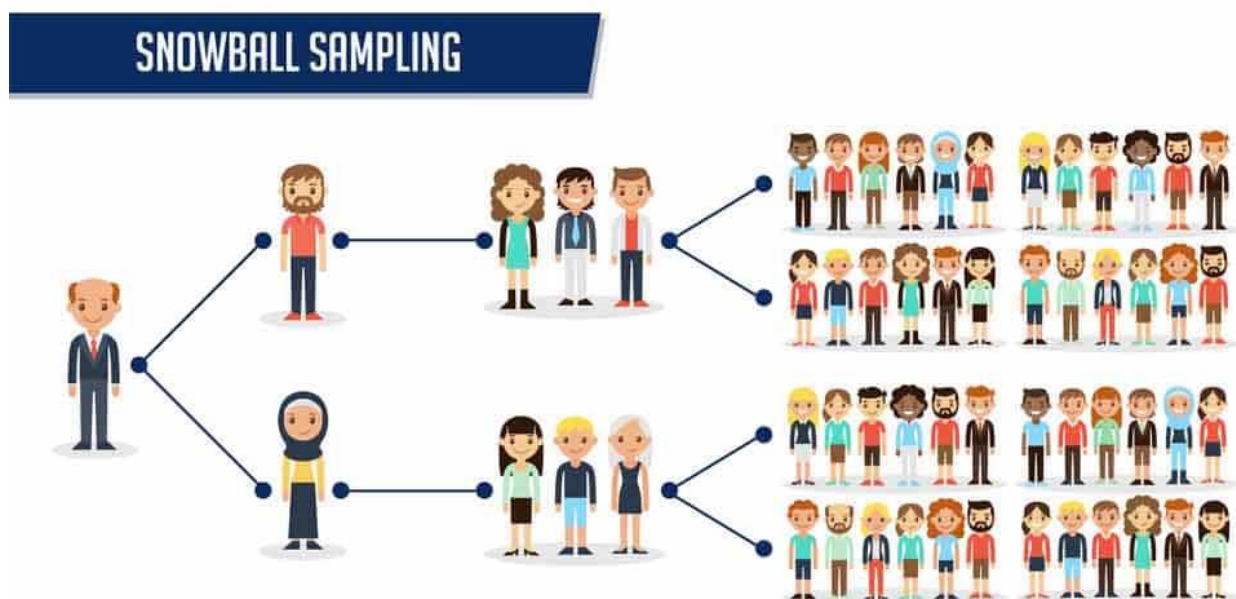
Here are two distinct Judgmental Sampling examples:

- Consider a scenario where a panel decides to understand what are the factors which lead a person to select ethical hacking as a profession. Ethical hacking is a skill which has been recently attracting youth. More and more people are selecting it as a profession. The researchers who understand what ethical hacking is will be able to decide who should form the sample to learn about it as a profession. That is when judgmental sampling is implemented. Researchers can easily filter out those participants who can be eligible to be a part of the research sample.
- There are many tribes in the world which have their own religious beliefs, for instance, the Balinese people follow syncretism, which is considered to be a mixture of Hinduism and Buddhism. For researchers who plan to study the culture of Southeast Asian countries, it is advised that they select strata using judgmental sampling as religious beliefs are considered to be highly sensitive in this part of the world. Due to the sensitivity of the topic, if samples of those who have appropriate knowledge are created and research is conducted with those samples, results will be highly accurate. Probability sampling techniques often produce altered results in such cases.

Judgmental Sampling Advantages

- **Consumes minimum time for execution:** In this sampling bias approach, researcher expertise is important and there are no other barriers involved due to which selecting a sample becomes extremely convenient.
- **Allows researchers to approach their target market directly:** There are no criteria involved in selecting a sample except for the researcher's preferences. Due to this, he/she can communicate directly with the target audience of their choice and produce desired results.
- **Almost real-time results:** A quick poll or survey can be conducted with the sample using judgmental sampling since the members of the sample will possess appropriate knowledge and understanding of the subject.

Snowball Sampling: Definition



Snowball sampling or chain-referral sampling is defined as a non-probability sampling technique in which the samples have rare traits. This is a sampling technique, in which existing subjects provide referrals to recruit samples required for a research study.

For example, if you are studying the level of customer satisfaction among the members of an elite country club, you will find it extremely difficult to collect primary data sources unless a member of the club agrees to have a direct conversation with you and provides the contact details of the other members of the club.

This sampling method involves a primary data source nominating other potential data sources that will be able to participate in the research studies. Snowball sampling method is purely based on referrals and that is how a researcher is able to generate a sample. Therefore this method is also called the chain-referral sampling method.

Snowball sampling is a popular business study method. The snowball sampling method is extensively used where a population is unknown and rare and it is tough to choose subjects to assemble them as samples for research.

This sampling technique can go on and on, just like a snowball increasing in size (in this case the sample size) till the time a researcher has enough data to analyze, to draw conclusive results that can help an organization make informed decisions.

Types of Snowball Sampling

1. **Linear Snowball Sampling:** The formation of a sample group starts with one individual subject providing information about just one other subject and then the chain continues with only one referral from one subject. This pattern is continued until enough number of subjects are available for the sample.
2. **Exponential Non-Discriminative Snowball Sampling:** In this type, the first subject is recruited and then he/she provides multiple referrals. Each new referral then provides with more data for referral and so on, until there is enough number of subjects for the sample.
3. **Exponential Discriminative Snowball Sampling:** In this technique, each subject gives multiple referrals, however, only one subject is recruited from each referral. The choice of a new subject depends on the nature of the research study.

Snowball Sampling Method

The nature of snowball sampling is such, that it cannot be considered for a representative sample or in that case for statistical studies. However, this sampling technique can be extensively used for conducting qualitative research, with a population that is hard to locate. Let us now explore how snowball sampling can be carried out:

Consider hypothetically, you as a researcher are studying the homeless in Texas City. It is obviously difficult to find a list of all the details of the number of homeless there. However, you are able to identify one or two homeless individuals who are willing to participate in your research studies.

Now, these homeless individuals provide you with the details of other homeless individuals they know. The first homeless individual that you found for your research is the primary data. You can collect the information and tabulate data from the primary data source and move on to other individuals who the primary data source has referred to. You as a researcher can continue to tap as many homeless you can find through the reference provided till you know you have collected enough data for your research.

The same strategy can be followed to conduct research or study individuals belonging to certain underground subculture, or individuals who have a hidden identity or are members of a cult etc. who don't want to be identified easily. Trust is an important part of any researcher.

An individual, who is ready to share information, needs to know that the information will be used discreetly and this kind of trust is especially important in snowball sampling. For a participant to agree to identify themselves or their group, researchers first need to develop that kind of rapport with the participants. Please know that this sampling technique may consume more time than anticipated because of its nature.

Snowball Sampling Applications

Snowball sampling is usually used in cases where there is no precalculated list of target population details (homeless people), there is immense pain involved in contacting members of the target population (victims of rare diseases), members of the target population are not inclined towards contributing due to a social stigma attached to them (hate-crime, rape or sexual abuse victims, sexuality, etc.) or the confidentiality of the organization respondents work for (CIA, FBI or terrorist organization).

Thus, this type of sampling is preferred in the following applications:

- **Medical Practices:** There are many less-researched diseases. There may be a restricted number of individuals suffering from diseases such as progeria, porphyria, Alice in Wonderland syndrome etc. Using snowball sampling, researchers can get in touch with these hard to contact sufferers and convince them to participate in the survey research.
- **Social research:** Social research is a field which requires as many participants as possible as it is a process where scientists learn about their target sample. When social research is to be conducted in domains where participants might not necessarily be willing to contribute such as homeless or the less-fortunate people.
- **Cases of discord:** In case of disputes such as an act of terrorism, violation of civil rights and other similar situations, the individuals involved may oppose giving their statements for evidential purposes. The researchers or management can use snowball sampling, to filter out those people from a population who are most likely to have caused the situation or are witness to the event to gather proof around the event.

Snowball Sampling Examples

For some population, snowball sampling is the only way of collecting data and meaningful information. Following are the instances, where snowball sampling can be used:

1. **No official list of names of the members:** This sampling technique can be used for a population, where there is no easily available data like their demographic information. For example, homeless or list of members of an elite club, whose personal details cannot be obtained easily.

2. **Difficulty to locate people:** People with rare diseases are quite difficult to locate. However, if a researcher is carrying out a research study similar in nature, finding the primary data source can be a challenge. Once he/she is identified, they usually have information about more such similar individuals.
3. **People who are not willing to be identified:** If a researcher is carrying out a study which involves collecting information/data from sex workers or victims of sexual assault or individuals who don't want to disclose their sexual orientations, these individuals will fall under this category.
4. **Secretiveness about their identity:** People who belong to a cult or are religious extremists or hackers usually fall under this category. A researcher will have to use snowball sampling to identify these individuals and extract information from them.

Advantages of Snowball Sampling

1. **It's quicker to find samples:** Referrals make it easy and quick to find subjects as they come from reliable sources. An additional task is saved for a researcher, this time can be used in conducting the study.
2. **Cost effective:** This method is cost effective as the referrals are obtained from a primary data source. It's is convenient and not so expensive as compared to other methods.
3. **Sample hesitant subjects:** Some people do not want to come forward and participate in research studies, because they don't want their identity to be exposed. Snowball sampling helps for this situation as they ask for a reference from people known to each other. There are some sections of the target population which are hard to contact. For example, if a researcher intends to understand the difficulties faced by HIV patients, other sampling methods will not be able to provide these sensitive samples. In snowball sampling, researchers can closely examine and filter members of a population infected by HIV and conduct a research by talking to them, making them understand the research objective, and eventually, analyzing the received feedback.

Disadvantages of Snowball Sampling

1. **Sampling bias and margin of error:** Since people refer those whom they know and have similar traits, this sampling method can have a potential sampling errors and margin of error. This means a researcher might only be able to reach out to a small group of people and may not be able to complete the study with conclusive results.
2. **Lack of cooperation:** There are fair chances even after referrals, people might not be cooperative and refuse to participate in the research studies.

Quota sampling:

Definition

Quota sampling is defined as a non-probability sampling method in which researchers create a convenience sample involving individuals that represent a population. Researchers choose these individuals according to specific traits or qualities. They decide and create quotas so that the market research samples can be useful in collecting data. These samples can be generalized to the entire population. The final subset will be decided only according to the interviewer's or researcher's knowledge of the population.

For example, a cigarette company wants to find out what age group prefers what brand of cigarettes in a particular city. They apply survey quota on the age groups of 21-30, 31-40, 41-50, and 51+. From this information, the researcher gauges the smoking trend among the population of the city.

Types of quota sampling:

Quota sampling can be of two kinds – controlled quota sampling and uncontrolled quota sampling. Here's what they mean:

Controlled quota sampling:

Controlled quota sampling imposes restrictions on the researcher's choice of samples. Here, the researcher is limited to the selection of samples.

Uncontrolled quota sampling:

Uncontrolled quota sampling does not impose any restrictions on the researcher's choice of samples. Here, the researcher chooses sample members at will.

Quota sampling example:

Let's look at a basic example of quota sampling:

A researcher wants to survey individuals about what smartphone brand they prefer to use. He/she considers a sample size of 500 respondents. Also, he/she is only interested in surveying ten states in the US. Here's how the researcher can divide the population by quotas:

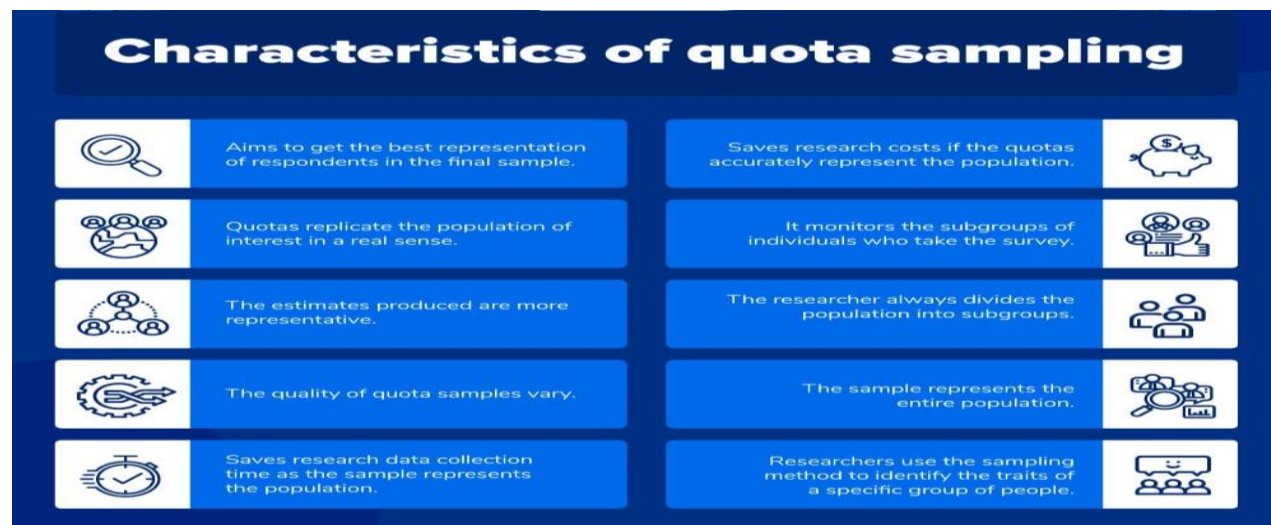
- Gender: 250 males and 250 females
- Age: 100 respondents each between the ages of 16-20, 21-30, 31-40, 41-50, and 51+
- Employment status: 350 employed and 150 unemployed people.
 - (Researchers apply further nested quotas . For eg, out of the 150 unemployed people, 100 must be students.)
- Location: 50 responses per state

Depending on the type of research, the researcher can apply quotas based on the sampling frame. It is not necessary for the researcher to divide the quotas equally. He/she divides the quotas as per his/her need (as shown in the example where the researcher interviews 350 employed and only 150 unemployed individuals). Random sampling can be conducted to reach out to the respondents.

How to perform quota sampling:

Probability sampling techniques involve a significant amount of rules that the researcher needs to follow to form samples. But, since quota sampling is a non-probability sampling technique, there are no rules for formally creating samples. Usually, there are four steps to form a quota sample. Here are the steps:

1. **Divide the sample population into subgroups:** With stratified random sampling, the researcher bifurcates the entire population into mutually exhaustive subgroups, i.e., the elements of each of the subgroups becomes a part of only one of those subgroups. Here, the researcher applies random selection.
2. **Figure out the weightage of subgroups:** The researcher evaluates the proportion in which the subgroups exist in the population. He/she maintains this proportion in the sample selected using this type of sampling method.
3. For example, if 58% of the people who are interested in purchasing your Bluetooth headphones are between the age group of 25-35 years, your subgroups also should have the same percentages of people belonging to the respective age group.
4. **Select an appropriate sample size:** In the third step, the researcher should select the sample size while maintaining the proportion evaluated in the previous step. If the population size is 500, the researcher can pick a sample of 50 elements. The sample chosen after following the first three steps should represent the target population.
5. **Conduct surveys according to the quotas defined:** Make sure to stick to the predefined quotas to achieve actual actionable results. Don't survey quotas that are full and focus on completing surveys for each quota.



Characteristics of quota sampling:

Here are the top ten characteristics of quota sampling

1. Aims to get the best representation of respondents in the final sample.
2. Quotas replicate the population of interest in a real sense.
3. The estimates produced are more representative.
4. The quality of quota samples vary.
5. Saves research data collection time as the sample represents the population.
6. Saves research costs if the quotas accurately represent the population.
7. It monitors the number of types of individuals who take the survey.
8. The researcher always divides the population into subgroups.
9. The sample represents the entire population.
10. Researchers use the sampling method to identify the traits of a specific group of people.

Advantages of quota sampling

Here are the top four advantages of quota sampling

1. **Saves time:** Because of the involvement of a quota for sample creation, this sampling process is quick and straightforward.
2. **Research convenience:** By using quota sampling and appropriate research questions, interpreting information and responses to the survey is a much convenient process for a researcher.
3. **Accurate representation of the population of interest:** Researchers effectively represent a population using this sampling technique. There is no room for over-representation as this sampling technique helps researchers to study the population using specific quotas.
4. **Saves money:** The budget required for executing this sampling method is minimalistic.

Applications of quota sampling:

Below are the instances where quota sampling is applied and used.

- In situations where researchers have specific criteria for conducting research, it allows the selection of subgroups, due to which it becomes extremely convenient for researchers to obtain desired results. A trait or characteristic can be the filter for subgroup formation.
- The researcher uses this method when he/she has time constraints. Applying quotas gives the researcher an idea of the whole population of interest in very little time.
- Quotas are applied when the researcher is on a tight budget. Instead of researching a large population, the researcher saves money by using a few quotas to get the whole picture of the population.
- Some research studies do not require pinpoint accuracy due to the nature of the research project. It is ideal for applying to quota sampling for these studies.

Errors in sampling.

Assume you are a market researcher of a company looking to introduce a new product to the market. You must collect data from a sample of potential customers as part of your research to determine their preferences and purchasing behavior. But how can you be sure that the information you get from your sample is accurate for all of the people who might buy your product? The idea of sampling error comes into play here.

It is the difference between what a sample has and what the entire population has. It can significantly affect how accurate and reliable market research data is.

We will cover ways to reduce sampling error in this article to get more accurate and reliable results. Now, grab your favorite coffee and get ready to explore what sampling error is.

What is a sampling error?

A sampling error occurs when the sample used in the study does not represent the entire population. Although sampling errors occur frequently, researchers always include a margin of error in their conclusions as a matter of statistical practice.

The margin of error is the amount allowed for a miscalculation to represent the difference between the sample and the actual population.

Sampling is a type of analysis where a small sample of observations is chosen from a larger population. The selection process can produce both sampling errors and non-sampling errors.

Recognizing sampling error

Sampling errors are differences between a sample's values and the actual population's values. This is because a sample does not accurately represent the whole population of data.

Since there was a mistake in collecting the data, the results from sampling are no longer valid. Also, when a sample is chosen randomly or because of bias, it doesn't represent the whole population, and sampling errors are likely to happen.

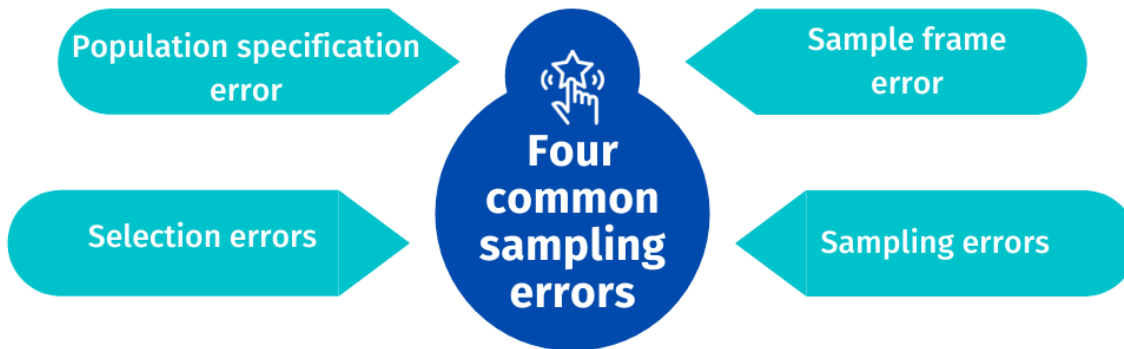
It is possible to avoid them if analysts carefully choose representative subsets of data from which to draw conclusions about the entire population. Factors including sample size and design, diversity in the population, and sampling percentage all contribute to sampling errors.

Diversity in the population increases the error in the estimations since it causes sampling to produce mixed results. Increasing the size of the samples allows them to represent the population more accurately, reducing the impact of population variation.

It is important to consider sampling errors before reporting survey results to establish confidence in the reliability of the estimates and their conclusions.

Types of common sampling error in market research

Here are the top four market research errors while sampling:



- **Population specification error**

A population specification error occurs when researchers don't know precisely who to survey.

For example, imagine a research study about kid's apparel. Who is the right person to survey? It can be both parents, only the mother, or the child. The parents make purchase decisions, but the kids may influence their choice.

- **Sample frame error**

Sampling frame error occurs when researchers target the sub-population wrongly while selecting the sample.

For example, picking a sampling frame from the telephone white pages book may have erroneous inclusions because people shift their cities. Erroneous exclusions occur when people prefer to un-list their numbers. Wealthy households may have more than one connection, thus leading to multiple inclusions.

- **Selection error**

Selection error occurs when respondents self-select themselves to participate in the study. You can control selection errors by going the extra step to request responses from the entire sample. Only interested ones respond.

Pre-survey planning, follow-ups, and a neat and clean survey design will boost respondents' participation rate. Also, try sampling methods like CATI surveys and in-person interviews to maximize responses.

- **Sampling errors**

Sampling errors occur due to a disparity in the representativeness of the respondents. It majorly happens when the researcher does not plan his sample carefully.

These sampling errors can be controlled and eliminated by creating a careful sample design, having a large enough sample to reflect the entire population, or using an online sample or survey audiences to collect responses.

Example of sampling error

Let's look more closely at this example.

Let's say a political party conducts a survey to find out how well-liked their candidate is before a major election. Instead of picking a random sample of the whole population to survey, they only ask their own members.

The sample would be skewed since party members may have extremely different opinions and tastes from the rest of the population. Party members may care more about their candidate's ideals or be more loyal to them. This may cause the survey to suggest more support than the general population has.

Suppose the survey results are used to make campaign decisions, such as where to allocate money or which issues to prioritize. In that case, they may not accurately reflect the candidate's support among the people. This could result in a poor campaign plan, affecting their election chances.

To avoid these kinds of sampling errors, it is essential to use a sampling method that is representative of the population being studied, such as random sampling or stratified random sampling, and to make sure that the sample size is big enough to give accurate results.

Controlling your sampling error

Statistical theories help researchers measure the probability of sampling errors in sample size and population. The size of the sample considered from the population primarily determines the size of the sampling error. Larger sample sizes tend to encounter a lower rate of errors.

Researchers use a metric known as the margin of error to understand and evaluate the margin of error. Usually, a confidence level of 95% is considered to be the desired confidence level.

Sampling error vs. Non-sampling error

Surveys can have sampling and non-sampling errors. Survey findings can be affected by sampling and non-sampling mistakes.

Sampling error arises when a survey sample does not accurately represent the population being researched due to random sampling. Sampling and non-response bias, measurement error, and sampling variability can cause this.

Non-sampling error includes all survey errors other than sampling errors. This includes questionnaire design, coding, data input, data collection, processing, and analysis errors. Improper interviewer training, insufficient or inaccurate data, or data analysis or reporting errors can create non-sampling errors.

Non-sampling error can be decreased by applying quality control measures and ensuring that all components of the survey process are properly designed, implemented, and monitored, whereas sampling error can be reduced by utilizing appropriate sampling procedures and increasing the sample size.

Sampling error vs. Sampling bias

Sampling in statistics means choosing the research group. Sampling error and sampling bias affect sample accuracy and representativeness in statistics.

Sampling error occurs because a sample is a subset of the population and may not precisely represent it. Instead, sampling bias occurs when the sample is not representative of the population. This may happen if the method used to choose the sample favors or excludes particular sorts of people, resulting in an overrepresentation or underrepresentation of particular groups.

Using stratified or random sampling and unbiased, population-representative sample selection can reduce sampling bias. On the other hand, sampling error can be reduced by using the proper sampling methods and making the sample size bigger.

What are the steps to reduce sampling errors?

Sampling errors are easy to identify. Here are a few simple steps to reduce sampling error:



- **Increase sample size**

A larger sample size is more accurate because the study gets closer to the actual population size.

- **Divide the population into groups**

Test groups according to their size in the population instead of a random sample. For example, if people of a specific demographic make up 20% of the population, make sure that your study is made up of this variable to reduce sampling bias.

- **Know your population**

Study your population and understand its demographic mix. Know what demographics use your product and service and ensure you only target the sample that matters.

We have also created a tool to help you determine your sample size easily: [Sample Size Calculator](#).

A sampling error is measurable, and researchers can use it to their advantage to estimate their findings' accuracy and variance.
